

MODULE 7 · COMPLETE

Foundational Model Fine-Tuning Module Summary & Bridge

Module 7 — Complete Concept Map

Sub-Module	Key Concepts	Key Methods
7.1 Fundamentals	What is FT, Full-param FT, FT vs Pre-training, Catastrophic forgetting, Transfer learning	HuggingFace Trainer, gradient checkpointing
7.2 PEFT	LoRA, QLoRA, Prefix Tuning, Adapter Modules, PEFT trade-offs	PEFT library, BitsAndBytes, AdapterHub
7.3 Instruction IT	Instruction datasets, Data quality, Multi-task IT, LIMA finding	Self-Instruct, ChatML, MinHash dedup
7.4 RLHF	RLHF pipeline, Reward model, Policy opt., PPO, DPO, Goodhart's Law	TRL: RewardTrainer, PPOTrainer, DPOTrainer
7.5 Data & Eval	Data collection, Preprocessing, Evaluation metrics	datasketch, Perspective API, LLM-as-judge

Module 7 — By the Numbers

- Fine-tuning is the customisation engine that bridges using foundation models as-is (Modules 1–5) and building them from scratch (Module 9)
- The central tension of the module: PEFT makes fine-tuning accessible; RLHF/DPO makes it safe and aligned
- Key insight: almost all of the world knowledge is in the pre-trained weights — fine-tuning teaches the model how to express it
- Production path: Base Model → QLoRA SFT → DPO alignment → HuggingFace Hub → LLMOps (Module 9)

What Unlocks Next — Module 8 & 9 Bridge

Module 8: AI Safety & Guardrails

- Evaluation frameworks for safety and alignment
- Guardrails: LLM output filtering (Llama Guard, NeMo)
- Red teaming and adversarial testing
- RLHF alignment from a safety perspective
- Prerequisite: Module 7 RLHF & DPO concepts

Module 9: LLMOps

- Deploying fine-tuned models at scale (vLLM, TGI)
- Model serving: batching, quantisation, speculative decoding
- Fine-tuned adapter management in production
- Monitoring: perplexity drift, output quality degradation
- Prerequisite: Module 7 fine-tuning pipeline

KEY REFERENCES

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Next: Module 8: AI Safety, Evaluation & Responsible Deployment